

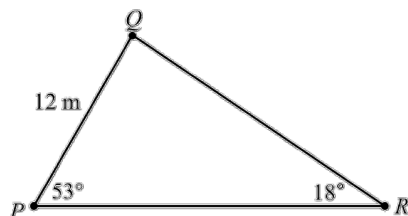
Non-Right Angled Trigonometry

Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 What is the distance from Q to R in the diagram below, correct to the nearest metre? 2



$$\frac{QR}{\sin 53^\circ} = \frac{12}{\sin 18^\circ} \quad (\text{Sine rule})$$

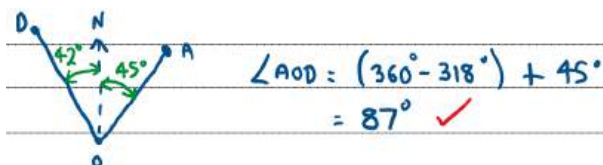
$$QR = \frac{12 \sin 53^\circ}{\sin 18^\circ}$$

$$= 31.013...$$

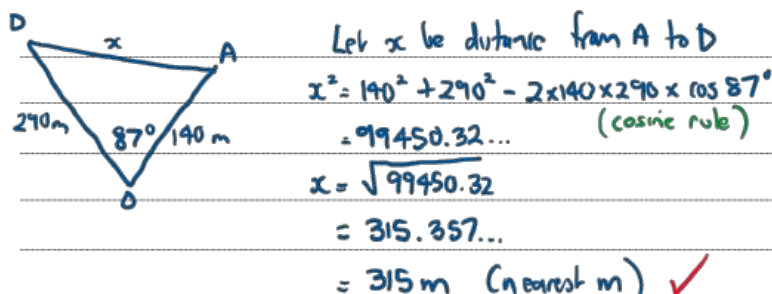
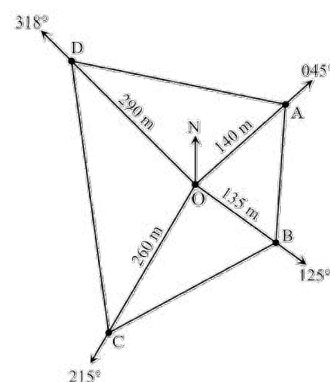
$$= 31 \text{ m (nearest m)}$$

- 2 A compass radial survey for block of land ABCD, is shown in the diagram below.

- a. Calculate the size of angle AOD. 1



- b. Calculate the distance along the boundary of the land from A to D, correct to the nearest metre. 2



- 3 What is the correct expression for x in triangle ABC? 1

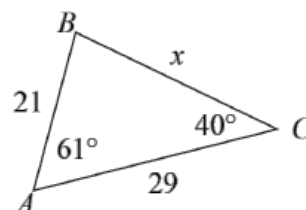
A. $x = \frac{29 \sin 79^\circ}{\sin 61^\circ}$

B. $x = \frac{21 \sin 40^\circ}{\sin 61^\circ}$

C. $x = \frac{29 \sin 61^\circ}{\sin 79^\circ}$

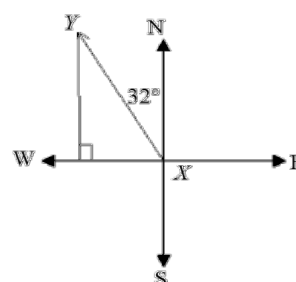
D. $x = \frac{21 \sin 40^\circ}{\sin 79^\circ}$

C $x = \frac{29 \sin 61^\circ}{\sin 79^\circ}$

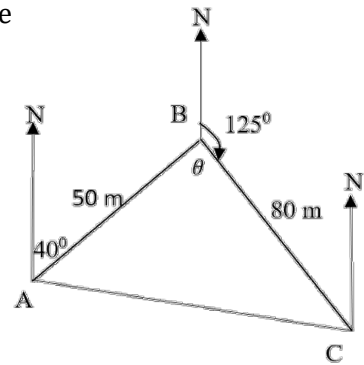


- 4 The compass bearing of Y from X is N32°W. What is the compass bearing of X from Y? 1

S32°E



- 5 The diagram below shows the plan for a triangular park ABC . Fence posts have been located at points A, B and C . AB is 50 m and BC is 80 m. The bearing of B from A is 040° and the bearing of C from B is 125° .



- a. What is the size of angle θ ?

$$\theta = 360 - (180 - 40) - 125 = 95^\circ$$

- b. What is the true bearing of point A from point B ?

$$\text{Bearing} = 125^\circ + 95^\circ = 220^\circ$$

Bearing of A from B is 220° .

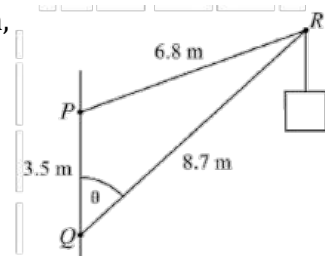
- c. Find the area of the park? Answer correct to one decimal place.

$$\begin{aligned} A &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} \times 50 \times 80 \times \sin 95^\circ \\ &= 1992.389... \approx 1992.4 \text{ m}^2 \\ \text{Area of the garden is } 1992.4 \text{ m}^2. \end{aligned}$$

- d. Find the length of AC ? Answer correct to one decimal place.

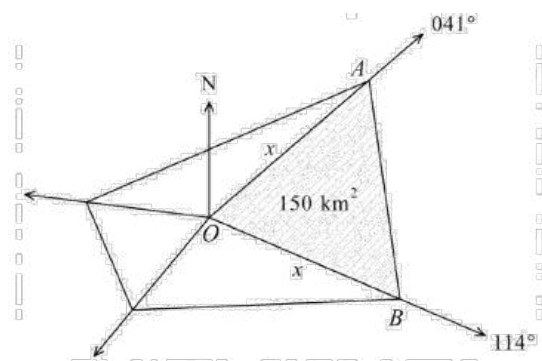
$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos A \\ AC^2 &= 50^2 + 80^2 - 2 \times 50 \times 80 \times \cos 95^\circ \\ AC &= 97.6553... \approx 97.7 \text{ m} \\ \text{Length of } AC &\text{ is } 97.7 \text{ m}. \end{aligned}$$

- 6 A jib crane is shown in the diagram below. In the diagram, $RQ = 8.7$ m, $PQ = 3.5$ m and $PR = 6.8$ m. Determine the angle PQR , correct to the nearest minute.



$$\begin{aligned} \cos \angle PQR &= \frac{8.7^2 + 3.5^2 - 6.8^2}{2(3.5)(8.7)} = 0.6847... \\ \angle PQR &= \cos^{-1}(0.6847...) = 46^\circ 47' \end{aligned}$$

- 7 A radial survey is taken on a plot of land. Point A and B are equal distance (x) in kilometres from O . Point A is on a bearing of 041° and point B is on a bearing of 114° . The area of triangle AOB is 150 km^2 . What is the distance x , in kilometres, correct to one decimal place?



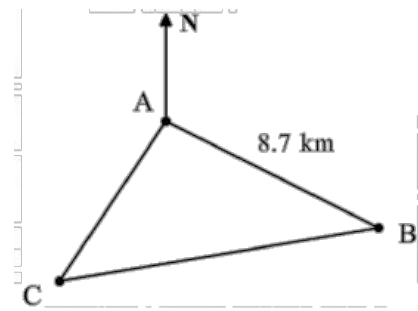
$$\begin{aligned} \frac{1}{2} x^2 \sin(114 - 41) &= 150 \\ x^2 &= \frac{300}{\sin 73^\circ} \\ &= 313.707... \\ x &= 17.711... = 17.7 \text{ km} \end{aligned}$$

- 8 There are 3 campsites, A, B and C, each connected by roads.

The length of the road between A and B is 8.7 km.

The following information is known:

- The bearing of B from A is 125°
- The bearing of C from A is 209°
- The bearing of B from C is 085°



- a. Calculate the length of the road between B and C, correct to 4 significant figures. 3

$$\angle CAB = 209 - 125 = 84^\circ$$

Bearing of A from C is 29°

$$\angle ACB = 85 - 29 = 56^\circ$$

$$\frac{BC}{\sin 84^\circ} = \frac{8.7}{\sin 56^\circ}$$

$$BC = 8.7 \times \frac{\sin 84^\circ}{\sin 56^\circ} = 10.44 \text{ km}$$

- b. Hence, or otherwise, find the area of the triangle enclosed by the 3 roads, correct to the nearest square metre. 2

$$\angle ABC = 180 - 84 - 56 = 40^\circ$$

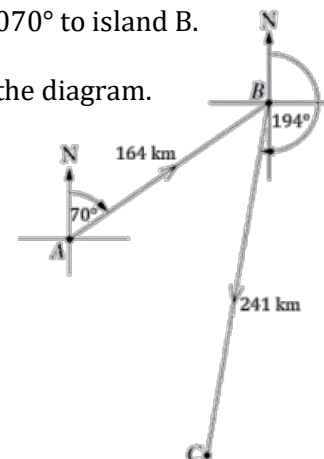
$$\text{Area} = \frac{1}{2} (10.44 \times 8.7) \sin 40^\circ = 29.19 \dots \text{km}^2 = 29191556 \text{ m}^2$$

- 9 Manavdeep leaves island A in a boat and sails 164 km on a bearing of 070° to island B.

He then sails on a bearing of 194° for 241 km to island C, as shown in the diagram.

- a. Show that the size of $\angle ABC = 56^\circ$. 1

$$\begin{aligned} 194 - 180 &= 14^\circ \\ 70^\circ - 14^\circ &= 56^\circ \end{aligned}$$



- b. Using the cosine rule, find the distance from A to C. Give your answer correct to the nearest whole number. 1

$$\begin{aligned} AC &= \sqrt{164^2 + 241^2 - 2 \times 164 \times 241 \times \cos 56^\circ} \\ &= 201.9255 \approx 202 \end{aligned}$$

- c. Using the sine rule, calculate $\angle BCA$ to the nearest degree. 2

$$\begin{aligned} \frac{\sin A}{164} &= \frac{\sin 56^\circ}{202} \\ A &= \sin^{-1} \left(\frac{164 \times \sin 56^\circ}{202} \right) = 42.305 \\ &= 42^\circ 18' 18.82'' \approx 42^\circ \end{aligned}$$

- d. What is the bearing of A from C to the nearest degree? 2

$$360^\circ - (42^\circ - 14^\circ) = 332^\circ$$

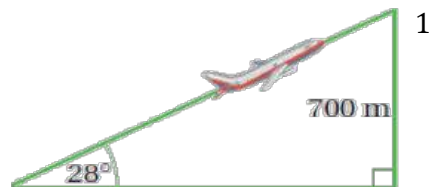
Non-Right Angled Trigonometry

Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

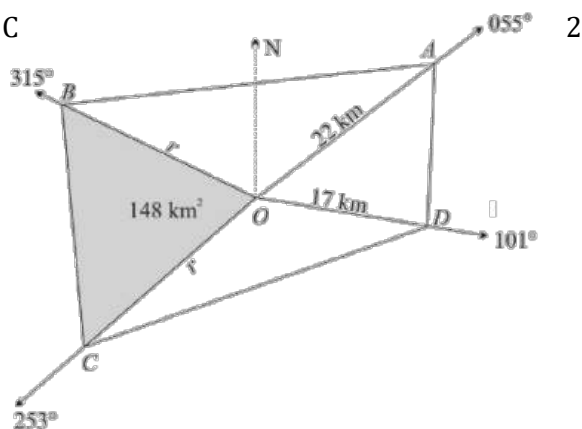
- 1 An aeroplane climbs at an angle of 28° to the horizontal. Find to the nearest metre the distance travelled along its flight path while rising 700m vertically. Express your answer to the nearest metre.



$$\sin 28^\circ = \frac{700}{x}$$

$$x = 1491.038 \approx 1491$$

- 2 A radial survey is taken on a plot of land. Points B and C are equal distance in kilometres from O. Point B is on a bearing of 315° and point C is on a bearing of 253° . The area of BOC is 148 km^2 .



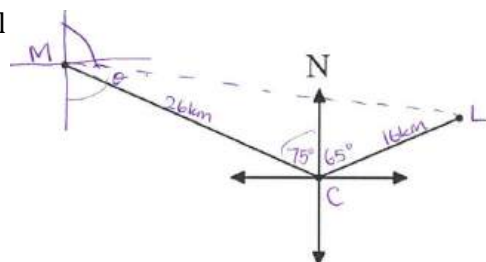
What is the distance OB, correct to one decimal place?

$$148 = \frac{1}{2} \times r^2 \times \sin 62$$

$$r^2 = 335.24$$

$$r = 18.3 \text{ km}$$

- 3 Two hikers decide to leave a campsite (C) to visit some natural landmarks in the area. One hiker walks 16 km on a bearing of 65° to see a lake (L); the other hiker walks 26 km on a bearing of 285° to reach the top of a mountain (M).



- a. Complete the diagram to display the scenario. Include lengths and bearings from the question above. 1

- b. Calculate the distance between the mountain and the lake, correct to the nearest kilometre. 1

$$c^2 = 26^2 + 16^2 - 2 \times 26 \times 16 \times \cos 140^\circ$$

$$= 1569.34 = 39.615 \approx 40 \text{ km}$$

- c. Hence, using your answer from part b) and the sine rule, find the bearing of the lake from the mountain, correct to the nearest degree. 2

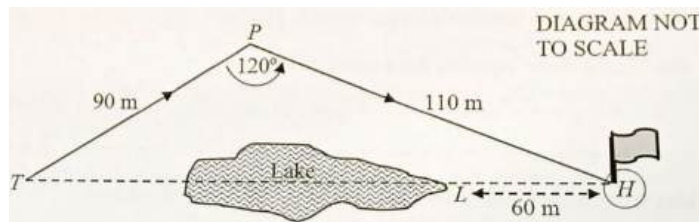
$$\frac{\sin \theta}{16} = \frac{\sin 140}{40}$$

$$\sin \theta = \frac{16 \sin 140}{40}$$

$$\theta = 14.89$$

$$\text{Bearing} = 180 - 75 - \theta \approx 90^\circ$$

- 4 From the "T" position on a golf course, the hole (H) on the green is 60 m beyond the furthest edge L of a lake and due east of T, as shown in the diagram. The point P on the golf course is such that angle $TPH = 120^\circ$ with $TP = 90$ m and $PH = 110$ m.



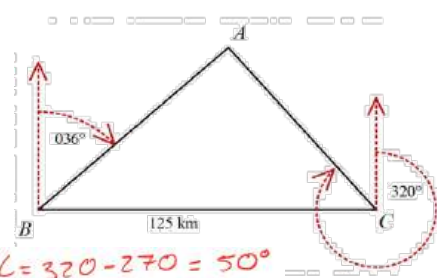
Use the Cosine rule to determine how far a golf ball must fly east from T to land at L.

$$c^2 = 90^2 + 110^2 - 2 \times 90 \times 110 \times \cos 120 = 30100$$

$$c = 173.49$$

$$TL = 173.49 - 60 = 113.49 \approx 113.5 \text{ m}$$

- 5 Town B is 125 km due west of town C. Town A is on a bearing 036° from B and on a bearing 320° from C. Which calculation could be used to find the distance from A to B?



A. $AB = \frac{125 \sin 36^\circ}{\sin 320^\circ}$

B. $AB = \frac{125 \sin 50^\circ}{\sin 54^\circ}$

C. $AB = \frac{125 \sin 54^\circ}{\sin 76^\circ}$

D. $AB = \frac{125 \sin 50^\circ}{\sin 76^\circ}$

$$\angle C = 320 - 270 = 50^\circ$$

$$\angle B = 90 - 36 = 54^\circ$$

$$\angle A = 180 - 50 - 54 = 76^\circ$$

$$\frac{AB}{\sin 50} = \frac{125}{\sin 76}$$

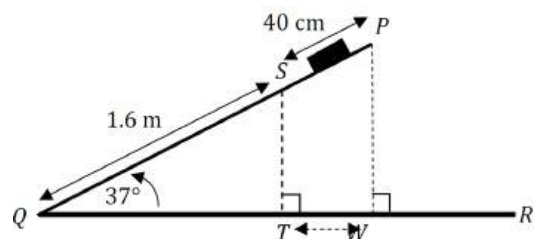
$$AB = \frac{125 \sin 50}{\sin 76}$$

- 6 Find the area of an equilateral triangle that has 4 cm sides.

$$A = \frac{1}{2} \times 4 \times 4 \times \sin 60$$

$$= 6.928 \text{ cm}^2$$

- 7 A coin is slid down a ramp PQ which is at an angle of elevation of 37° to a horizontal bench QR. When the coin reaches the point S on the ramp 40 cm from P, it has 1.6 m to slide to reach Q.



- a. Use Trigonometry to calculate the distance QT correct to TWO decimal places.

$$\cos 37 = \frac{QT}{1.6}$$

$$QT = 1.6 \cos 37 = 1.28 \text{ m}$$

- b. What is the horizontal distance TW in the diagram, correct to 1 decimal place?

using the fact that similar triangles have sides in proportion

$$\frac{2}{1.6} = \frac{QW}{1.28}$$

$$QW = 1.6$$

$$TW = 1.6 - 1.28 = 0.32 = 0.3$$

- 8 In the diagram X, Y and Z represent the locations of three towns.

The town Y is due east of X, and the bearing of Z from Y is 046° .

- a. Find the size of $\angle XYZ$.

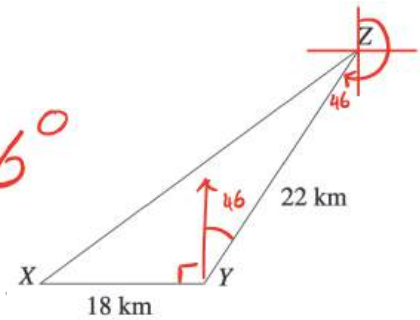
$$90 + 46 = 136^\circ$$

- b. Find the distance XZ correct to one decimal place.

$$XZ^2 = 18^2 + 22^2 - 2 \times 18 \times 22 \times \cos 136$$

$$XZ^2 = 1377.72$$

$$XZ = 37.1 \text{ km}$$

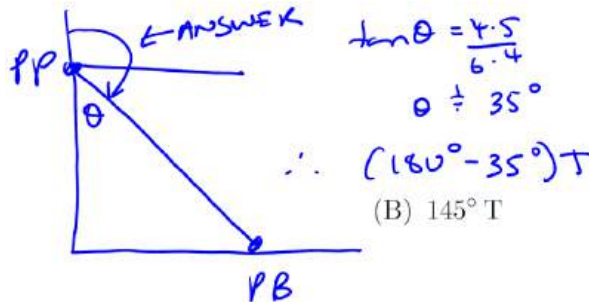


- c. What is the bearing of Y from Z?

$$180 + 46 = 226^\circ$$

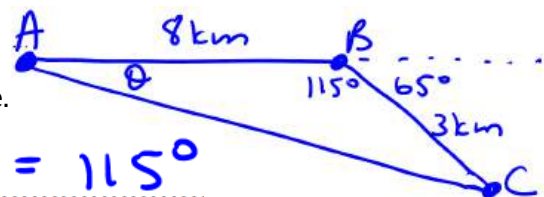
- 9 Picnic Point is 4.5 km west and 6.4 km north of Padstow Bay. What is the bearing of Padstow Bay from Picnic Point?

Bay from Picnic Point?



- 10 A bushwalker starts at point A and walks 8 km in the direction due east to point B. At B he turns and walks on a bearing of 155° T for 3 km to C.

- a. Draw a diagram illustrating the situation above.



- b. Find the value of $\angle ABC$.

$$\angle ABC = 115^\circ$$

- c. Calculate the distance AC. Give your answer correct to one decimal place.

$$AC^2 = 8^2 + 3^2 - 2(8)(3)\cos 115^\circ$$

$$\therefore AC \approx 9.7 \text{ km}$$

- d. Calculate the bearing of C from A. Give your answer correct

to the nearest degree.

$$\frac{\sin \theta}{3} = \frac{\sin 115^\circ}{9.7}$$

$$\theta \approx 16^\circ$$

$$\therefore \text{Bearing is } 106^\circ \text{ T}$$

- 11 The information from a compass radial survey of a field is shown in the diagram above.

- a. Find the perimeter of the field. Give your answer correct to

the nearest metre.

$$AB^2 = 48^2 + 54^2 - 2(48)(54)\cos 110^\circ$$

$$AB = 83.62$$

$$BC^2 = 54^2 + 36^2 - 2(54)(36)\cos 76^\circ$$

$$BC = 57.20$$

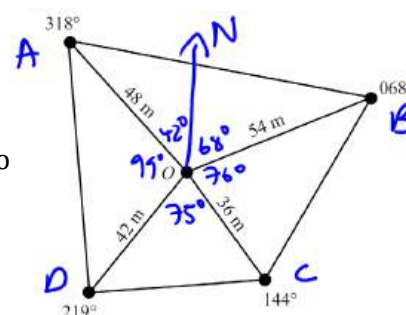
$$DC^2 = 36^2 + 42^2 - 2(36)(42)\cos 75^\circ$$

$$DC = 47.72$$

$$AD^2 = 48^2 + 42^2 - 2(48)(42)\cos 99^\circ$$

$$AD = 68.55$$

$$\therefore \text{Perimeter is approximately } 257 \text{ m}$$



b. Find the area of the field. Give your answer correct to the nearest square metre.

2

$$\begin{aligned} A &= \frac{1}{2}(48)(54)\sin 110^\circ + \frac{1}{2}(54)(36)\sin 76^\circ \\ &+ \frac{1}{2}(36)(42)\sin 75^\circ + \frac{1}{2}(48)(42)\sin 90^\circ \\ &\doteq 3887 \text{ m}^2 \quad \checkmark \end{aligned}$$